

Sara Asghari

✉ sara2asghari@gmail.com | ☎ 778 862 7378 | in SaraAsghari | 🌐 S-Asghari

Technical Skills

Languages/DevOps: Python, C++, C, SQL, Bash Scripting, Git, Docker, CI/CD, AWS (EC2, RDS, IAM), GCP
AI/LLM: PyTorch, Scikit-Learn, Pandas, NumPy, Prompt Engineering, RAG, LangChain, LlamaIndex, HuggingFace
Backend/Databases: FastAPI, Flask, Node.js, Next.js, PySpark, MySQL, PostgreSQL, MongoDB, BigQuery

Work Experience

Avicenna Technologies | Calgary, AB, CA | Remote
AI/ML Engineer

Apr 2025 - Present

Designed and deployed production AI infrastructure for a mobile health application, integrating LLM-based symptom assessment and personalized recommendations. Owned system architecture, API development, and cloud deployment.

- Architected an intelligent health agent on **Vertex AI**, applying **prompt engineering** and response optimization to deliver context-aware recommendations, and leveraging **function calling** to interface with user health records.
- Built a **Python** backend with **RESTful APIs** using **FastAPI** and **Docker**; deployed on **GCP Cloud Run** via **CI/CD** pipelines and fronted by an **API Gateway** to manage request routing and monitor service health.
- Implemented **unit and integration testing** using **pytest**, with automated execution in **CI/CD** pipelines to ensure reliable production deployments.
- Secured and managed user data by integrating **Firebase Authentication** for request-level validation and utilizing **MongoDB** for scalable storage and retrieval of user profiles and health survey data.

Farpoint Technologies | Vancouver, BC, CA
AI/ML Engineer

Mar 2024 - Oct 2024

Developed a full-stack AI-powered web application to automate meeting documentation, task management, and document retrieval, accelerating internal workflows and reducing employee workload by 50%.

- Built a multi-service backend using **Flask** and **Node.js**; containerized with **Docker**, deployed on **AWS EC2**, and integrated with **PostgreSQL** on **Amazon RDS**; implemented **CI/CD** pipelines using **GitHub Actions**.
- Engineered a **RAG AI agent** using **LlamaIndex** and **PGVector** to index and query document embeddings across Google Drive and Microsoft Graph APIs via **OAuth 2.0**, storing embeddings in **PostgreSQL**.
- Developed a type-safe **Next.js** frontend in **TypeScript**. Leveraged the App Router to implement client-side data synchronization and facilitate asynchronous communication with **RESTful endpoints**.
- Collaborated in an **agile** setting, contributing to sprint planning, issue tracking, and coordination using **Jira**.

University of Victoria | Victoria, BC, CA
Graduate Research Assistant

Jan 2023 - Nov 2024

Developed LLM-based data pipelines and computational frameworks for automated fact verification, emphasizing performance, correctness, and computational efficiency.

- Built inference infrastructure by **fine-tuning** and deploying **large language models**, applying quantization, GPU acceleration, and tokenization strategies to improve throughput and reduce infrastructure costs.
- Designed a verification system architecture that translates unstructured claims into first-order logic to perform structured validation against large-scale textual datasets—achieving up to a 76% relative improvement in F1-MacroAvg over existing baselines while maintaining high system reliability.

Institute for Research in Fundamental Sciences (IPM) | Tehran, IR
Research Intern

July 2019 - Sep 2019

Analyzed neural (LFP) signals from monkeys using **feature extraction** (source/sink patterns, peaks, stimulus-response latency, firing rates). Applied PCA and t-SNE for **dimensionality reduction**, then evaluated K-Means and DBSCAN to identify the optimal **clustering** method for patterns tied to attention shifts and decision-making.

Notable Projects

Twitter Bot Detection via Network Analysis | NumPy, Pandas, NetworkX, Scikit-Learn, Keras

Identified distinguishing patterns of fake accounts by modeling user interactions as graphs and computing node-level metrics. Analyzed distributions and pairwise correlations to select features for ML-based bot detection.

Movie Recommender System | Python

Developed a recommendation engine to process and analyze over one million user interactions, implementing collaborative filtering (user-user, item-item) and baseline techniques to deliver personalized content.

Parallel Structural Clustering for Large-Scale Networks | MapReduce, PySpark

- Architected a distributed clustering system to identify structural patterns in massive network datasets with millions of nodes, leveraging parallel processing techniques.
- Implemented a **MapReduce**-based algorithm in **PySpark**, optimizing partition strategies and reducing data shuffling to efficiently process graph data at scale.

Education

University of Victoria

M.Sc. in Computer Science | GPA: 3.92/4.0

Jan 2023 - Nov 2024

Amirkabir University of Technology

B.Sc. in Computer Software Engineering | GPA: 3.77/4.0

Sep 2016 - Mar 2022